

## 34 Nitrogen compounds

### A · 2026 examination · Concept guide and original exam-style practice

#### Understand the concept

Amines act as bases and nucleophiles; amides are much less basic because the nitrogen lone pair delocalises into the carbonyl. Amino acids exist as zwitterions near their isoelectric point, affecting solubility and electrophoretic movement. Diazonium compounds can form azo dyes via coupling with activated aromatic rings under suitable cold conditions.

#### Worked example: follow the reasoning

In an amide, resonance lowers availability of N lone pair for protonation relative to an amine.

#### Quick check · multiple choice

A Which is more basic in water? [1]

- A Ethanamide
- B Ethanamine
- C Both equal by definition
- D Neither can protonate

Write your choice: \_\_\_\_\_

#### Structured practice

Answer in the space provided. Show working and units where relevant.

1 Compare basicity of ethanamine and ethanamide using their nitrogen lone pairs. [3]

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## Answers and explanation

Quick check: B. Ethanamine

### Question 1 [3 marks]

Ethanamine is more basic [1]; its N lone pair is available to accept  $\text{H}^+$  [1]; amide N lone pair delocalises into  $\text{C}=\text{O}$  and is less available [1].

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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